Curriculum Vitae

Dr. Tejaswini Kar

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Education

Kalinga Institute of Industrial Technology (KIIT) Deemed to be University, Bhubaneswar
Degree: Ph.D, 2018(**Awarded**).
Thesis Topic: Illumination Invariant Similarity measures for Video Shot Boundary detection

Kalinga Institute of Industrial Technology (KIIT) University, Bhubaneswar
Major: Communication System Engineering
Degree: M.Tech, 2008
Thesis Topic:Adaptive watermarking Robust to Geometric Distortion.

BPUT, Bhubaneswar
Major: Electronics & Telecommunication Engineering
Degree: B.Tech , July, 2003.

Ramadevi Women's College (Council of Higher Secondary Education)
Degree: Intermediate (12th), Science, 1998

Unit-4, Governement Girls High school (Higher Secondary Education)
Degree: 10th Class, 1996

Professional Appointments

1. Worked as a Lecturer in College of Engineering , Bhubaneswar from June 2004 to July 2008.

2. Worked as Lecturer in Kalinga Institute of Industrial Technology , Deemed to be University, Bhubaneswar from July 2008 to September 2009.

3. Worked as Assistant Professor(I) in Kalinga Institute of Industrial Technology , Deemed to be University, Bhubaneswar from July 2009 to September 2017.

4. Worked as an Assistant Professor(II) in Kalinga Institute of Industrial Technology , Deemed to be University, Bhubaneswar from July 2017 to 31st September 2023.

5. Worked as an Associate Professor in Kalinga Institute of Industrial Technology , Deemed to be University, Bhubaneswar from October 2023 to till today.

6. Organizing committee member in 3rd international conference on Communication circuits and systems held at School of Electronics Engineering, KIIT, Bhubaneswar, India from 16th to 18th October 2020.

7. Organising Committee member AESPC 2023, held at School of Electronics Engineering, KIIT, Bhubaneswar, India

Honors and Awards

1. Received the **'Certificate of Excellence'** award for SAMSUNG-PRISM project awarded by SAMSUNG in 2022 along with Cash Award.

2. Won the "**International Research Awards on New Science Inventions"** under the category of **"Women Researcher Award", NESIN 2021 (**Entry ID: **10583).**

3. **Received best paper award** in the international conference ICDMAI 2019 held in **Malaysia** during 18th to 20th January 2019.

4. Presented a poster in National science day Poster Presentation in 2014 held in KIIT.

FDP/STTP Attended

1) Successfully completed "**UHV-1"** in online mode from 22nd May to 26th May 2023, Organised by AICTE.

2) Successfully completed "**UHV-2"** in online mode from 16th October to 21st October 2023 Organised by AICTE.

3) Participated in workshop **"VLSI to system Design: Silicon to End Application Approach"** from 31st July to 4th August 2023.

4) Completed " **Neural Network and Deep Learning**" certification course from course Era in 2023.

5) Successfully participated in the event, "**3rd One Week International Workshop on Signal Processing using Machine Learning: Applications & Hands-on (Online)**", organized by School of Electronics Engineering (SENSE), VIT-AP University, Amaravati (AP), India and technically sponsored by MathWorks, IEEE Guntur Subsection, and IEEE VIT-AP Student Branch during 05th October to 11th October 2023.

6) Participated in Five Days FDP on **"IPR Awareness and Patent Prosecution"** organised by Department of Electronics and Communication, Oriental Institute of Science and Technology, Bhopal during 13th - 17th July 2021.

7) Participation in the AICTE-QIP-STC on "Advanced **Applications in Signal Processing and Artificial Intelligence**" conducted by the School of Electrical Sciences, IIT Bhubaneswar from 6th July to 10th July 2021.

8) Participated and successfully completed One Week Online Faculty Development Program on **Emerging Trends in Machine Learning & Signal Processing** Organized by KIIT Deemed to be University, Bhubaneswar, Odisha, form 20th to 26th June, 2021.

9) Participated & completed successfully AICTE Training And Learning (ATAL) Academy Online FDP on "**Machine Learning and Applications**" from 1-10-2020 to 5-10-2020 at Kalinga Institute of Industrial Technology.

10) Participated & completed successfully AICTE Training And Learning (ATAL) Academy **Online** FDP on "**Augmented Reality (AR)/ Virtual Reality (VR)**" from 2-11-2020 to 6-11-2020 at C.V. Raman Global University.

11) Participated & completed successfully AICTE Training And Learning (ATAL) Academy Online FDP on "**Internet of Things (IoT)**" from 2020-12-7 to 2020-12-11 at Indian Institute of Information Technology Allahabad.

12) Participated in FDP on "**Python and emerging Trends in Machine Learning**", conducted by Forsk coding school, held between 2nd to 6th June 2020.

13) Participated in the Faculty Development Programme on "**Deep Learning for Medical Data Analysis**" held from 20-12-2020 to 24-12-2020 at "Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur".

14) Participated in skill enhancement programme on "**Deep Learning**" conducted by Forsk Coding school, held between 9th-13th June 2020.

15) Participated in the AICTE Sponsored One Week Online Short Term Training Programme (STTP) On **"SciLab: An Effective Open Source Tool for Computer Vision and Machine Learning"** conducted by the Department of Electronics and Communication Engineering, Vidya Jyothi Institute of Technology (Autonomous) in association with AICTE from 14-12-2020 to 19-12-2020.

16) Participated in TEQIP-III sponsored online Faculty Development Programme on "**Signal Processing Applications**" held from 21st to 25th Sept20 organized by Department of Instrumentation & Electronics Engineering, College of Engineering and Technology (Autonomous), Bhubaneswar, Odisha.

17) Participated in the National Workshop on **"Image and Video Processing**", (NWIVP- 2018) organized by the Department of Electronics and Communication Engineering, Silicon Institute of Technology, Bhubaneswar held on 28th and 29th September 2018.

18) Participated in Faculty Development Program on **"FICE (Intel IoT)"** conducted by Intel and Organized by School of Electronics Engineering, KIIT University, Bhubaneswar During June 23-30, 2017.

19) Participated in Faculty Development Program on **"LabVIEW in Signal Processing and Communications**" From 27th June to 30th June 2016, Conducted by National Instruments, India in KIIT University, Bhubaneswar.

20) Participated in Short-Term Course on "**Advanced Image Processing Solutions for Emerging Challenges in Medical Applications**" (AIPSECMA-2016) 25-26-September 2016, organized by Department of Electrical Engineering. National lnstitute of Technology, Rourkela.

Patents

Sl.no	Patent	Inventors Name	Institutions Name	Current Status
1	**Smart Mask Shaped Mouthset Capable of Enhancing and Synthesizing Speech Even From Lip Movement (Indian Patent)**	Dr. M. R. Arun, Prof.(Dr.) Bhagirathi Nayak, Tejaswini Kar, Dr. Patteti Krishna, Dr. Pradeep Kumar, R. Poonguzhali, Kavitha. T, Dr. Mauparna Nandan, Dr. Pramod V. R., Dr. Munish Jindal, Dr. Sangeetha, Dr. D. Vanathi, A. Arshath Kumar		Published

2	**A system and a method for Improving Current Profile in a Hybrid Energy Source (Australian Patent)**	Dr. T. Kar	KIIT Deemed to be University	Granted
3	**Smart Healthcare system for monitoring and control of the indoor environment using IOT technology and artificial Intelligence (Australian Patent)**	Dr. T. Kar	KIIT Deemed to be University	Granted
4	**IOT BASED, SMART AIR POLLUTION MONITORING AND CONTROLLING SYSTEM (Indian Patent)**	Dr. T. Kar	KIIT deemed to be University	Published
5	Wearable Textile antenna for telemedicine Application (Indian Design Patent)	Dr. T. Kar	KIIT Deemed to be University	Design Registered

Publications

Refereed Journal Publications

1. Kar, T., Kanungo, P., Mohanty, S.N. et al. Video shot-boundary detection: issues, challenges and solutions. Artif Intell Rev. 57, 104 (2024). https://doi.org/10.1007/s10462-024-10742-1. *(SCI)*

2. P Priyadarshini, P Kanungo, T Kar, Multigrade Brain tumor classification in MRI images using Fine tuned EfficientNet, e-Prime-Advances in Electrical Engineering, Electronics and Energy, 100498. (SCOPUS)

3. Anshuman Sharma, Biswaroop Nath, Tejaswini Kar, D Khasim, GANs for Face Completion: A Novel Deep Learning Approach , Vol. 9 (2023): EAI Endorsed Transactions on Pervasive Health and Technology. *(SCI)*

4. Tapaswini Pattnaik, Priyadarshi Kanungo, Tejaswini Kar, Prabodh Kumar Sahoo, Multiple linear regression based illumination normalization for non-uniform light image thresholding, e-Prime - Advances in Electrical Engineering, Electronics and Energy Vol. 7, (Elsevier, March 2024) Article number 100411.

5. S. Acharya, T. Kar, U. C. Samal, P. K. Patra,Performance Comparison between SVM and LS-SVM for Rice Leaf Disease detection, EAI Endorsed Transactions on Scalable Information Systems, Vol. 10, Issue 6, 2023.

6. D. Nayak, K.B. Ray, **T. Kar**, Chiman Kawan, A novel saliency based image compression algorithm using low complexity block truncation coding, Multimedia tools and application 82(30), pp. 47367-47385 (Springer 2023)

7. D. Nayak, K. B. Ray, **T. Kar**, S. N. Mohanty, Fuzzy Rule Based Adaptive Block Compressive Sensing for WSN Application, Mathematics, 11(7),1660 (MDPI **2023)**.

8. **T. Kar**, P. Kanungo, A gradient based dual detection model for shot boundary detection, Multimedia Tools and Applications, 82(6), pp. 8489-8506 (Springer,**2023**).

9. D. Nayak, K. Ray, **T. Kar**, C. Kwan, Walsh–Hadamard Kernel Feature-Based Image Compression Using DCT with Bi-Level Quantization. *Computers* **2022**, *11*, 110. https://doi.org/10.3390/computers11070110

10. D. Shome, **T. Kar** , S. N. Mohanty, et al. COVID-Transformer: Interpretable COVID-19 Detection Using Vision Transformer for Healthcare. *Int J Environ Res Public Health*. 2021;18(21):11086. Published Oct 21. doi:10.3390/ijerph182111086 *(SCI)*

11. **T. Kar**, P. Kanungo, "Motion and illumination defiant cut detection based on Weber features", IET Image Processing, 12(10), 1903-1912, October 2018.

12. **T. Kar**, P. Kanungo, "A motion and illumination resilient framework for automatic shot boundary detection", Signal, Image and Video Processing, 11(7), 1237-1244, October 2017.

13. **T. Kar**, P. Kanungo, “A Novel Method of Shot Boundary Detection using Centre Symmetric Local binary pattern”, IJERT 2016.

14. Vinod Kumar, Priya Nayan, Richa Nayan, **T. Kar**, “A decision and Repetition based algorithm for removing high density impulse noise”, IJERT 2016.

Conferences

1. S. Pal, **T. Kar**, P. Kanungo and T. Pattnaik, "Composer Identification in Classical Genre," 2023 IEEE 3rd International Conference on Applied Electromagnetics, Signal Processing, & Communication (AESPC), Bhubaneswar, India, 2023, pp. 1-6, doi: 10.1109/AESPC59761.2023.10390140.

2. **T. Kar**, S. Pahadsingh, N. C. Giri, O. Kuziakin, S. Leliuk and S. Bilyk, "Smart Plant Monitoring and Controlling System for Greenhouse Application Using IoT," 2023 IEEE 5th International Conference on Modern Electrical and Energy System (MEES), Kremenchuk, Ukraine, 2023, pp. 1-5, doi: 10.1109/MEES61502.2023.10402451.

3. **T. Kar**, S. Pahadsingh, N. C. Giri, M. Kharchenko, S. Leliuk and M. Khrypunov, "Smart LPG Gas Detection and Automatic Booking System for Home Safety Using IoT Platform," 2023 IEEE 4th KhPI Week on Advanced Technology (KhPIWeek), Kharkiv, Ukraine, 2023, pp. 1-5, doi: 10.1109/KhPIWeek61412.2023.10312969.

4. D. Nayak**,** K. B. Ray**, T. Kar**, “Performance Comparison of Different CS based Reconstruction Methods for WSN Application”, 2021 IEEE 2nd International Conference on Applied Electromagnetics, Signal Processing, & Communication (AESPC),DOI: 10.1109/AESPC52704.2021.9708523

5. Aurosweta Mahapatra, Sasmita Pahadsingh, **T. Kar**, “Transfer learning based COVID-19 detection Using Radiological Images”,2021 IEEE 2nd International Conference on Applied Electromagnetics, Signal Processing, & Communication (AESPC), 10.1109/AESPC52704.2021.9708473

6. D. Shome and **T. Kar**, "ConOffense: Multi-modal multitask Contrastive learning for offensive content identification," *2021 IEEE International Conference on Big Data (Big Data)*, 2021, pp. 4524-4529, doi: 10.1109/BigData52589.2021.9671427.

7. **T. Kar**, P. Kanungo and Vinod Jha, “Illumination Insensitive Video Cut Detection using Phase Congruency, Presented in (ICCTA 2021 Paper Presented).

8. D. Shome, **T. Kar**, “FedAffect: Few-shot federated learning for facial expression recognition”.(ICCV 2021),Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops, 2021, pp. 4168-4175.

9. S. Kattyayan**,** **T. Kar,** P Kanungo**, “**Performance Evaluation of Learning Based Frameworks for Devanagari Character Recognition**”,** 7th IEEE Uttar Pradesh Section International Conference on Electrical, Electronics and Computer Engineering, UPCON 2020.

10. **T. Kar**, P. Kanungo, “Abrupt scene change detection using block based Local directional Pattern”, ICDMAI 2019, Malaysia 18th to 20th January 2019, Advances in Intelligent Systems and Computing book series (AISC, volume 1016), Data Management, Analytics and Innovation pp 191-203.

11. **T. Kar**, P. Kanungo, "Video shot boundary detection based on Hilbert and wavelet transform" 2nd International Conference on Man and Machine Interfacing (MAMI), 2017, Bhubaneswar, India.

12. **T. Kar**, P. Kanungo, "Cut detection using local image descriptor", Electrical Computer and Electronics Engineering (UPCON) 2016 IEEE Uttar Pradesh Section International Conference, I.I.T. B.H.U., Varanasi, India pp. 378-383, 2016.

13. **T. Kar**, P. Kanungo,"A Novel Method of Shot Boundary Detection using Centre Symmetric Local Binary Pattern", IC3S 2016 conference proceedings (IJERT) 4(28),pp. 39-43.

14. **T. Kar**, P. Kanungo, Cut detection using block based center symmetric local binary pattern", 2015 International Conference on Man and Machine Interfacing (MAMI), pp. 1-5, December 2015, Bhubaneswar, India.

15. **T. Kar**, P. Kanungo, "A texture based method for scene change detection", 2015 IEEE Power Communication and Information Technology Conference (PCITC), pp. 72-77, October 2015, Bhubaneswar, India.

Book Chapter

1. Indrashis Mitra, Yashi Srivastava, Kananbala Ray, **Tejaswini Kar** "IoMT based Smart Health Monitoring: The Future of Health Care", "*Big data analytics in Fog-enabled IoT Networks: towards a privacy and Security perspective approach*". CRC Press.(***Accepted 2022***)

2. Nayak. D.**, Ray,** K. B., **Kar, T, "A Comparative Analysis of BTC Variants"**, Lecture Notes in Electrical Engineering, 2021, 728 LNEE, pp. 163–170 .

3. **Kar, T.,** Kanungo, P., Jha, V., Automatic Cut Detection: A Probability Based Approach Lecture Notes in Electrical Engineering, 2022, 894 LNEE, pp. 105–114.

4. **Kar, T.,** Kanungo, P., Jha, V. Illumination Insensitive Video Cut Detection Using Phase Congruency, Lecture Notes in Networks and Systems, 2022, 426, pp. 317–324.

Book Edited

Artificial Intelligence in Forecasting: Tools and Techniques. *CRC Press.*

Invited Speaker

1. Served as an **expert speaker** in AICTE Sponsored Six Days Short Term Training Program (STTP) On "**Machine Learning and Computer Vision**" **(Phase -I)**, held from 7th March to 12th March, 2022 organized by Computer Science & Engineering Department, Tripura Institute of Technology, Agartala, Tripura, India in online mode.

2. Served as an **Expert speaker** in AICTE Sponsored Six Days Short Term Training Program on "**Machine Learning and Computer Vision" (STTP on MLCV) (Phase -II)**, held from 21st March to 26th March, 2022 organized by Computer Science & Engineering Department, Tripura Institute of Technology, Agartala, Tripura, India in online mode.

3. Served as an **Expert speaker** in AICTE Sponsored Six Days Short Term Training Program on "**Machine Learning and Computer Vision" (STTP on MLCV) (Phase -III)** , held from 11th April to 16th April, 2022 organized by Computer Science & Engineering Department, Tripura Institute of Technology, Agartala, Tripura, India in online mode.

Membership in Professional Organizations

Life member of ISTE

Life member of ISCA

Senior Member IEEE (Signal Processing Society)

Research Students Guided (Ph.D.)

Name of the student, year	*Status*
Mrs. Tanu Trushna Mohapatra (Registered under KIIT)	Continuing
Mr Sandip Mishra (Registered Under KIIT)	Continuing
Mrs. Smaranika Bilas (Registered under KIIT)	Continuing

Editorial Board Member (*Scienctific Reports,Springer)*

Reviewer

- IET Image Processing
- ELCVIA Electronic Letters on Computer Vision and Image Analysis
- Int J. of Image & Graphics (IJIG)
- Journal of King Saud University of Science
- Hindawi
- Multimedia tools and Application (Springer)
- Scientific Reports

Scopus Id: 57189231239

Orcid Id: https://orcid.org/0000-0001-9020-3626

Web of Science ResearcherID: ABG-9331-2020

Subject Skills for Teaching on Theory	*Subject Skills for Teaching on Practical*
Signals and Networks	Matlab
Digital Signal Processing	Python
Digital Image Processing	
Soft Computing and its Application	
Cloud Computing	
Machine Learning	

Extracurricular activity

Coordinator and Resource person in Student Summer Internship Program on "**Machine Learning using Python**" form 17th May to 5th of June 2021, at KIIT Deemed to be University, Bhubaneswar, Odisha.

Career Objective

Seeking a challenging career with a progressive, result oriented organization which offers ample opportunity to prove, improve and grow in career and professional advancement.